A CASE-STUDY BASED REPORT

ON

Heart Disease Prediction Using Machine Learning: A Comparative Case Study of Logistic Regression, SVM and KNN Classifiers

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

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INDEX

Abstract 3

Introduction 4

Background 4
Motivation for the Project 4
Overview of the Work Carried Out 4
Scope and Responsible-Use Statement 5

Problem Statement 6

Case-Study Scenario 6
Central Case-Study Question 6
Formal Problem Definition 6
Objectives that Guide the Study 6
Assumptions and Limitations 7

Dataset and Exploratory Data Analysis 8

About the Dataset 8
Purpose of EDA 8
Preparation Steps Actually Performed 8
EDA Exploration - Implementation 9
Summary of EDA Findings 12

Algorithm Selection and Implementation 14

Conclusion 16

Abstract

Cardiovascular disease remains one of the leading causes of death worldwide, and early identification of high-risk patients can substantially improve outcomes. This case study investigates whether routinely collected clinical measurements can be used to predict the presence of heart disease with supervised machine learning. Using a publicly available heart-disease dataset of 303 patient records and 13 clinical attributes, the data was cleaned (one duplicate record removed, no missing values found), explored visually and statistically, split into 80% training and 20% testing sets using stratified sampling, and standardised with StandardScaler.

Three classification algorithms were implemented and compared: Logistic Regression, Support Vector Machine (RBF kernel) and K-Nearest Neighbours (k = 5). On the held-out test set of 61 patients, the SVM achieved the best results with an accuracy of 83.6%, precision of 79.5%, recall of 93.9% and F1-score of 0.861, followed by KNN (80.3% accuracy) and Logistic Regression (78.7% accuracy). The high recall of the SVM is particularly relevant in a medical-screening context, where missing a patient with the disease is more costly than raising a false alarm. A command-line prediction tool that combines the three models by majority vote was also built. The study is intended for learning purposes and is not a substitute for clinical diagnosis.

Keywords: heart disease prediction, machine learning, logistic regression, support vector machine, K-nearest neighbours, exploratory data analysis.

Introduction

Background

Heart disease covers a range of conditions that affect the heart's structure and function, including coronary artery disease, arrhythmias and heart failure. Doctors usually assess risk by combining several indicators such as age, sex, type of chest pain, resting blood pressure, cholesterol, electrocardiogram (ECG) results and the heart's response to exercise. Interpreting these indicators together is difficult because no single measurement is conclusive, and the relationships between them are often non-linear.

Machine learning (ML) offers a data-driven way of learning such combined patterns from historical patient records. In supervised classification, an algorithm is trained on records whose outcome is already known and then predicts the outcome for new, unseen patients. This makes ML a natural fit for decision-support applications in healthcare.

Motivation for the Project

The motivation for this project comes from three observations:

- Heart disease is a major cause of mortality, and early screening can lead to earlier treatment and lifestyle changes.
- Clinical data is already collected during routine check-ups, so a model built on it adds no extra cost or burden for the patient.
- The dataset is small and well-structured, which makes it ideal for understanding the complete ML workflow: cleaning, EDA, pre-processing, model training, evaluation and deployment of a simple predictor.

Overview of the Work Carried Out

The work was carried out in Google Colab using Python with pandas, NumPy, Matplotlib, Seaborn and scikit-learn. The main stages were:

- **Data cleaning:** checking the shape, missing values, duplicates and the valid values of categorical attributes; removing one duplicate record and saving the cleaned dataset.
- **Exploratory data analysis:** summary statistics, class balance, distributions, relationships between key features and the target, and a correlation heatmap.
- **Pre-processing:** separating features and target, a stratified 80/20 train-test split and feature scaling.

- **Modelling:** training Logistic Regression, SVM and KNN classifiers.
- **Evaluation:** accuracy, precision, recall, F1-score, classification reports and confusion matrices, followed by a side-by-side comparison.
- **Prediction system:** an interactive module that accepts a new patient's details and reports each model's prediction and a majority-vote final result.

Scope and Responsible-Use Statement

The scope of this study is limited to binary classification (heart disease present or absent) on a single, small, publicly available dataset using three classical ML algorithms with default or near-default settings. It does not include deep learning, hyper-parameter optimisation, external validation or any clinical trial.

Responsible use: the models in this report are built for academic learning only. They must not be used to diagnose, treat or make decisions about real patients. Predictions from such a model can be wrong, and any health concern should always be discussed with a qualified medical professional. The dataset is anonymised and publicly available, and no personally identifiable information was used.

Problem Statement

Case-Study Scenario

Consider a primary-care clinic that records a standard set of measurements for every adult who attends a cardiac check-up: age, sex, chest-pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate achieved during an exercise test, exercise-induced angina, ST depression (oldpeak), the slope of the peak-exercise ST segment, the number of major vessels seen on fluoroscopy and a thalassemia test result. The clinic has limited specialist time and would like an automated tool that flags patients who are likely to have heart disease so that they can be prioritised for further examination.

Central Case-Study Question

Can a machine learning model trained on routine clinical measurements reliably predict whether a patient has heart disease, and which of the three candidate algorithms (Logistic Regression, SVM or KNN) performs best for this task?

Formal Problem Definition

Let $X = (x_1, x_2, \ldots, x_{13})$ denote the vector of 13 clinical attributes of a patient and $y \in \{0, 1\}$ the diagnosis, where $y = 1$ means heart disease is present and $y = 0$ means it is absent. Given a training set of labelled records $\{(X_i, y_i)\}$, the task is to learn a function $f(X) \rightarrow \hat{y}$ that minimises prediction error on unseen patients, evaluated using accuracy, precision, recall and F1-score on a held-out test set.

Objectives that Guide the Study

- To clean and validate the heart-disease dataset and make it suitable for modelling.
- To explore the data and identify which clinical attributes are most associated with heart disease.
- To build and train Logistic Regression, SVM and KNN classifiers on scaled data.
- To evaluate the models with multiple metrics and confusion matrices, and select the best performer.
- To develop a simple interactive predictor that accepts a new patient's data and gives an ensemble prediction.

Assumptions and Limitations

- The dataset is assumed to be a representative and correctly labelled sample; the target value 1 is interpreted as heart disease present.
- The dataset is small (302 records after cleaning), so the test set contains only 61 patients. One patient changes the accuracy by about 1.6 percentage points, so the differences between models are indicative rather than conclusive.
- A single train-test split (random_state = 42) was used; cross-validation was not performed.
- Some categorical attributes contain values at the edge of or outside their documented ranges (for example ca = 4 and thal = 0); these were retained in this study.
- Models were trained with default or standard hyper-parameters and were not tuned.
- Numeric-coded categorical attributes (such as cp, restecg, slope, ca and thal) were scaled as numeric features rather than one-hot encoded.

Dataset and Exploratory Data Analysis

About the Dataset

The study uses a heart-disease dataset stored as heart.csv, derived from the well-known UCI heart-disease data and widely distributed for ML education. It originally contains 303 patient records and 14 columns: 13 input features and one binary target. After removing a duplicate record, 302 records remain. No missing values were present. Table 1 describes each attribute.

Table 1: Attributes of the heart-disease dataset

Attribute	Description	Type
age	Age of the patient in years	Numeric
sex	Sex (1 = male, 0 = female)	Binary
cp	Chest-pain type (0–3)	Categorical
trestbps	Resting blood pressure (mm Hg)	Numeric
chol	Serum cholesterol (mg/dl)	Numeric
fbs	Fasting blood sugar > 120 mg/dl (1 = yes, 0 = no)	Binary
restecg	Resting ECG result (0–2)	Categorical
thalach	Maximum heart rate achieved	Numeric
exang	Exercise-induced angina (1 = yes, 0 = no)	Binary
oldpeak	ST depression induced by exercise relative to rest	Numeric
slope	Slope of the peak-exercise ST segment (0–2)	Categorical
ca	Number of major vessels coloured by fluoroscopy (0–4)	Categorical
thal	Thalassemia test result (0–3)	Categorical
target	Diagnosis: 1 = heart disease present, 0 = absent	Target (binary)

The target is fairly balanced: 164 patients (54.3%) have heart disease and 138 patients (45.7%) do not. This means accuracy is a reasonable metric and no resampling technique is required.

Purpose of EDA

Exploratory Data Analysis (EDA) was performed before modelling to understand the structure and quality of the data, to detect anomalies and imbalance, to see how individual attributes relate to the diagnosis, and to detect strongly correlated features. These insights guide the choice of pre-processing steps and help interpret model behaviour afterwards.

Preparation Steps Actually Performed

- **Loaded the data** with pandas from heart.csv and inspected the first rows; the shape was (303, 14).

- **Missing values:** isnull().sum() showed zero missing values in every column.
- **Duplicates:** one duplicate row was found and removed with drop_duplicates(), leaving a shape of (302, 14); a second check confirmed zero duplicates.
- **Consistency check:** the unique values of sex, cp, fbs, restecg, exang, slope, ca, thal and target were listed and compared with their expected codes.
- **Saved the cleaned dataset** as heart_cleaned.csv after confirming with df.info() that all 14 columns are non-null numeric types.
- **Feature/target separation:** X contains the 13 features and y the target column.
- **Train-test split:** 80% training (241 records) and 20% testing (61 records) using train_test_split with random_state = 42 and stratify = y so that both sets keep the same class proportions.
- **Feature scaling:** StandardScaler was fitted on the training data only and then applied to both sets, which avoids data leakage.

EDA Exploration - Implementation

Descriptive statistics were first obtained with df.describe(). The average patient is about 54 years old (range 29–77), has a resting blood pressure of about 132 mm Hg, cholesterol of about 246 mg/dl (maximum 564, indicating a possible outlier) and a maximum heart rate of about 150 bpm (range 71–202). The mean oldpeak is about 1.04 with a maximum of 6.2. About 68% of the patients are male.

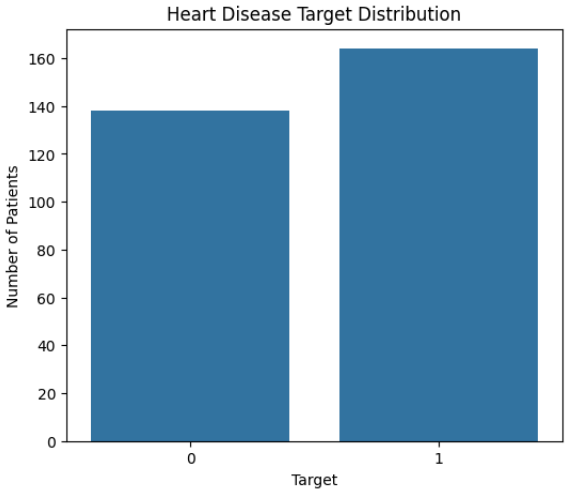


Figure 1: Distribution of the target (heart disease) classes

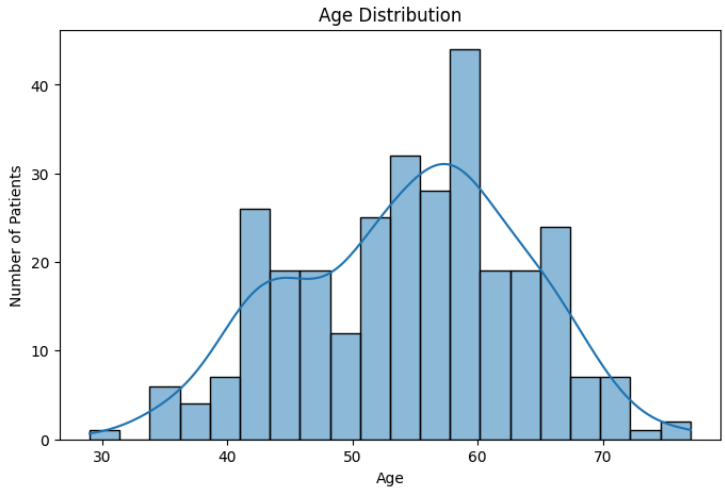


Figure 2: Age distribution of the patients

The age histogram shows that most patients are between about 45 and 65 years old, with a peak in the mid-50s to early 60s.

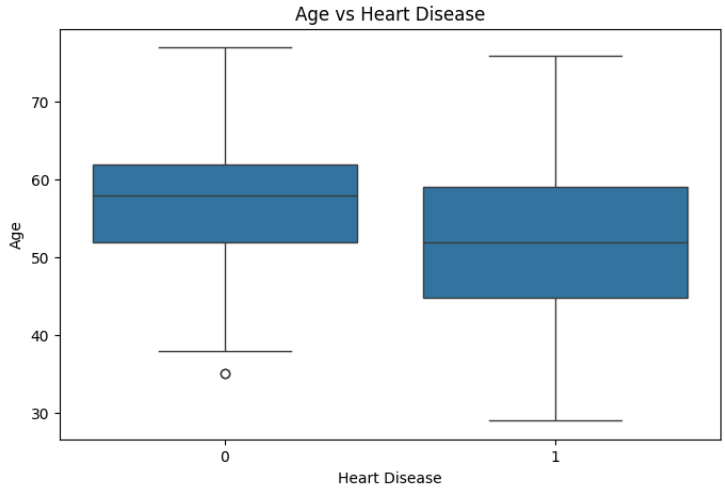


Figure 3: Age versus heart disease

Patients with heart disease tend to be somewhat younger on average than those without it, which is consistent with the weak negative correlation (−0.22) between age and the target in this dataset.

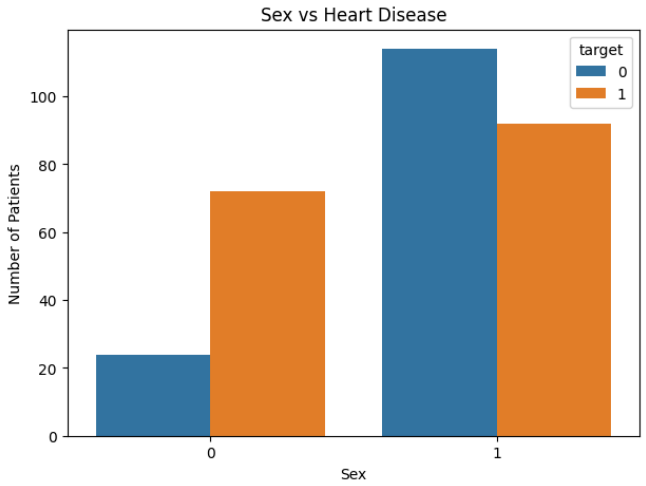


Figure 4: Sex versus heart disease (0 = female, 1 = male)

Roughly three quarters of the female patients (about 72 of 96) have heart disease, whereas among male patients the share is only about 45% (about 92 of 206). Note that this reflects how this particular sample was collected and should not be generalised.

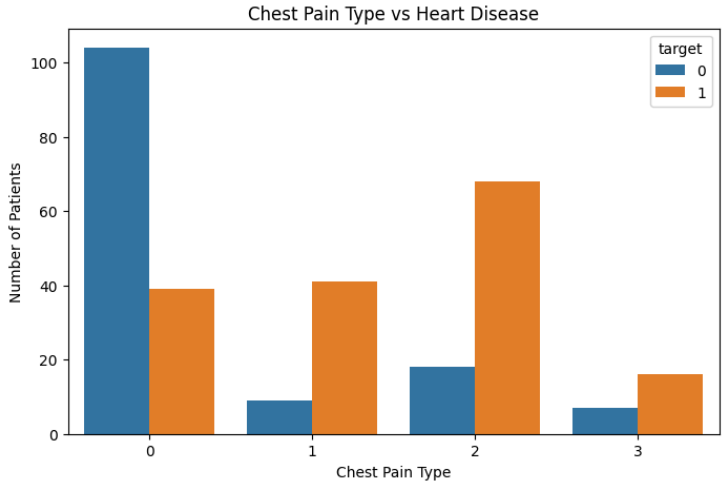


Figure 5: Chest-pain type versus heart disease

Patients with chest-pain type 0 are mostly healthy (about 104 of 143), while types 1, 2 and 3 are predominantly associated with heart disease. Chest-pain type is the strongest positive correlate of the target (0.43).

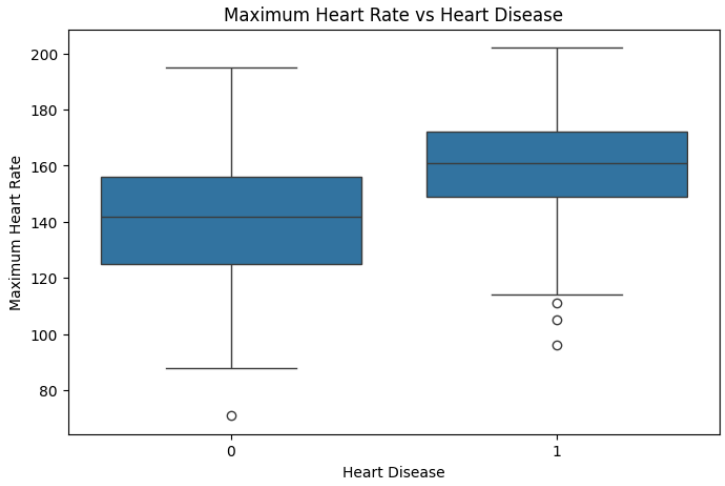


Figure 6: Maximum heart rate achieved versus heart disease

Patients with heart disease achieve a higher maximum heart rate on average than healthy patients, giving a positive correlation of 0.42.

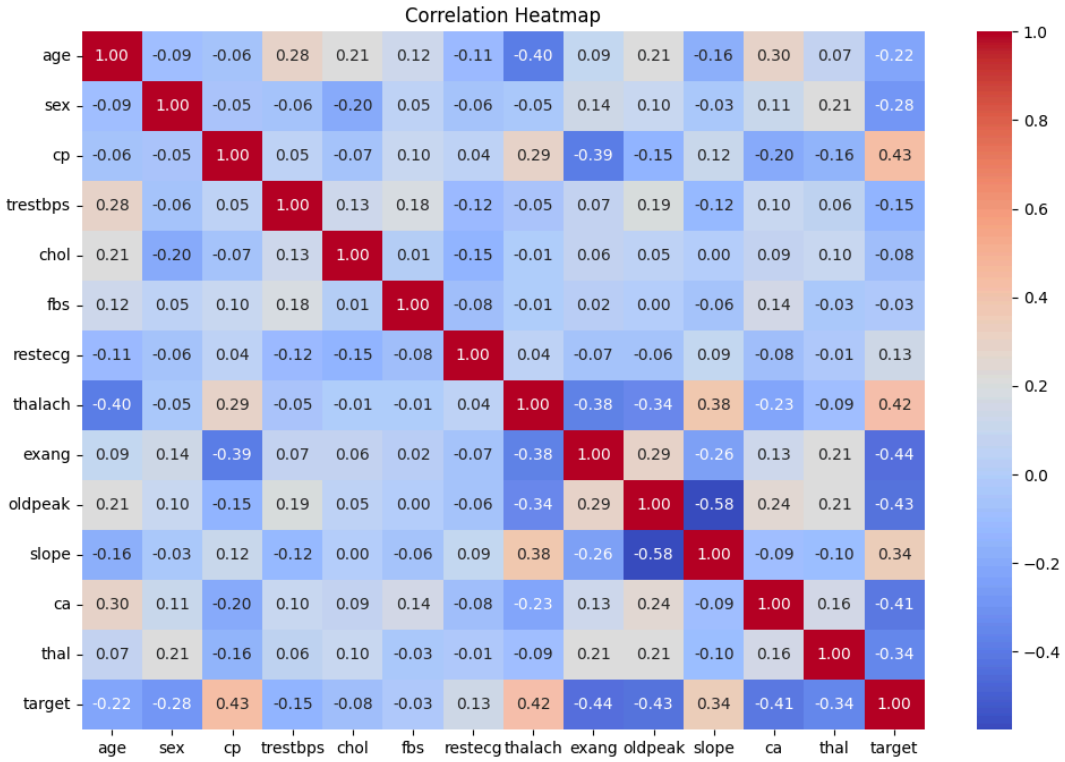


Figure 7: Correlation heatmap of all attributes

Summary of EDA Findings

- The data is clean: no missing values, only one duplicate, and a nearly balanced target (54.3% vs 45.7%).
- The attributes most positively correlated with heart disease are chest-pain type (0.43), maximum heart rate (0.42) and slope (0.34).

- The attributes most negatively correlated are exercise-induced angina (−0.44), oldpeak (−0.43), number of major vessels (−0.41) and thal (−0.34).
- Cholesterol (−0.08), fasting blood sugar (−0.03) and resting blood pressure (−0.15) show only weak linear relationships with the diagnosis.
- There is notable correlation between some features, for example oldpeak and slope (−0.58), age and maximum heart rate (−0.40), and chest-pain type and exercise-induced angina (−0.39). No pair is so strongly correlated that a feature must be dropped.
- Features are on very different scales (for example cholesterol up to 564 versus oldpeak up to 6.2), so scaling is essential for distance-based and margin-based models such as KNN and SVM.

Algorithm Selection and Implementation

The problem is a binary classification task on a small tabular dataset with 13 features. Three well-understood algorithms were selected because they represent different learning principles, which makes the comparison more informative:

- **Logistic Regression** is a linear model that estimates the probability of heart disease using a sigmoid function. It is fast, interpretable and a standard baseline for medical classification.
- **Support Vector Machine (SVM)** finds the maximum-margin boundary between the classes. With the RBF kernel it can model non-linear relationships and usually performs well on small, scaled datasets.
- **K-Nearest Neighbours (KNN)** classifies a patient according to the majority class among the k most similar patients in the training data. It is simple, non-parametric and makes no assumption about the data distribution.

Table 2: Model configuration used in the implementation

Model	scikit-learn class	Settings used
Logistic Regression	LogisticRegression	max_iter = 1000
SVM	SVC	kernel = 'rbf' (default C and gamma)
KNN	KNeighborsClassifier	n_neighbors = 5

Implementation. All three models were trained on the standardised training set (241 records) and then used to predict the 61 test records. Scaling was fitted only on the training data to prevent information from the test set leaking into the models. Performance was measured with accuracy, precision, recall and F1-score, where the positive class (1) is heart disease.

Table 3: Performance of the models on the test set (61 patients)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.787	0.763	0.879	0.817
SVM	**0.836**	**0.795**	**0.939**	**0.861**
KNN	0.803	0.769	0.909	0.833

The confusion matrices (Figure 8) correspond to the following counts on the test set of 33 diseased and 28 healthy patients, derived from the reported metrics: Logistic Regression correctly identified 29 of 33 diseased patients and 19 of 28 healthy patients; SVM identified 31 of 33 diseased and 20 of 28 healthy patients; and KNN identified 30 of 33 diseased and 19 of 28 healthy patients.

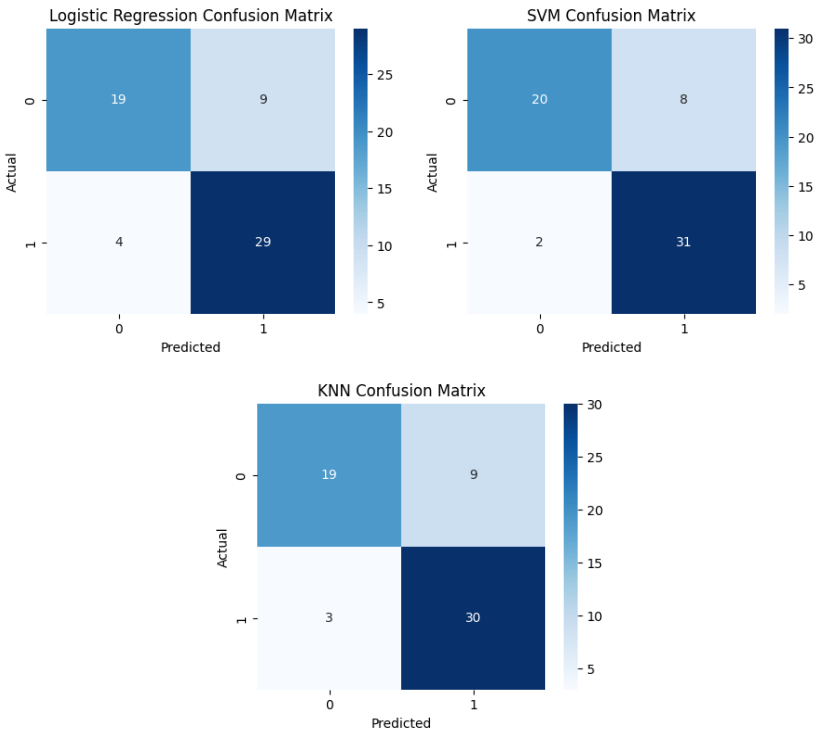


Figure 8: Confusion matrices of Logistic Regression (left), SVM (centre) and KNN (right)

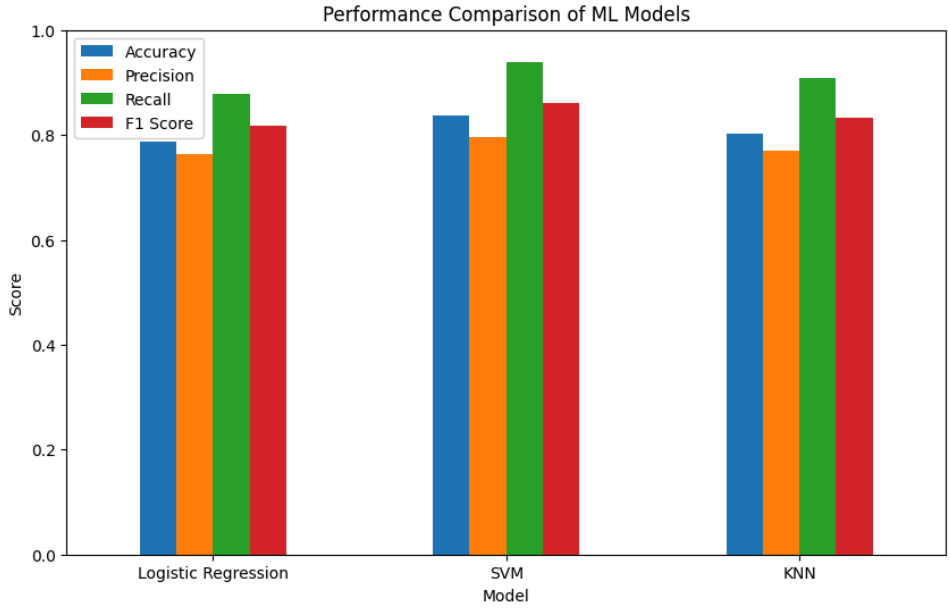


Figure 9: Performance comparison of the three models

Discussion of results. The SVM was the best model on every metric. It missed only 2 of the 33 diseased patients (recall 0.939) compared with 3 for KNN and 4 for Logistic Regression. All three models achieved higher recall than specificity, meaning they are better at catching sick patients than at correctly clearing healthy ones; the healthy class recall was about 0.68–0.71. In a screening setting this is an acceptable trade-off, because a false alarm leads to further tests

whereas a missed case may delay treatment. Because the test set is small, the margin between the models should be confirmed with cross-validation.

Prediction system. A final interactive module reads a new patient's 13 values, scales them with the saved StandardScaler, obtains a prediction from each of the three models and reports the majority vote (at least two of three models) as the final result. This ensemble approach reduces the risk of relying on a single model's error.

Conclusion

This case study demonstrated the complete machine learning workflow for predicting heart disease from clinical data: cleaning, EDA, pre-processing, training of three classifiers, evaluation and a simple prediction tool. The EDA showed that chest-pain type, maximum heart rate, exercise-induced angina, oldpeak, number of major vessels and thal are the attributes most associated with the diagnosis. Among the models tested, the SVM with an RBF kernel gave the best results (accuracy 83.6%, recall 93.9%, F1-score 0.861), followed by KNN and Logistic Regression.

The study has limitations: a small dataset, a single train-test split, untuned models, and some questionable category values that were retained. **Future work** can include k-fold cross-validation, hyper-parameter tuning with GridSearchCV, one-hot encoding of categorical attributes, handling the invalid values of ca and thal, adding ROC-AUC analysis and threshold tuning to favour recall, trying ensemble models such as Random Forest or Gradient Boosting, and validating on an independent dataset.

In conclusion, the results suggest that machine learning can provide useful decision support for heart-disease screening, but such models should only supplement, and never replace, the judgement of a qualified clinician.

Signature of the Team members